Engine Optimization, minification, obfuscation. Native and cross platform mobile application development. Desktop app development.

**SWE 4633 Network Security Credit: 3.00**

Basic Concepts: Security goals - confidentiality, integrity and availability, network security threats, security mechanisms, basics of cryptography;

Physical and Logical Access Control - Identification, Authentication and authorization - Windows and UNIX password system;

Mutual Authentication - Authentication protocols, Trusted Intermediaries, Mediated Authentication (with KDC), Many to many authentication, Kerberos Authentication requirements, Authentication functions, Message authentication codes;

Network Level Security Controls: Network layer security - IP security (IPSec), Transport Layer Security TLS/SSL, Electronic mail security - PGP, S/MIME, Web security, VPN and Real time Communication Security, Multimedia security (SRTP and MIKey);

System Level Security Controls: Intrusion detection, Password management, Malware, Anti malware, Firewall and its design principles, Intrusion Detection System, Trusted systems.

**Recommended Texts:**

1. J. M. Kizza, *Guide to Computer Network Security*, 6th ed. Springer, 2024
2. W. Stallings, *Network Security Essentials: Applications and Standards*, 6th ed. Pearson, 2016

**SWE 4635 Computer Graphics and Multimedia Credit: 3.00**

Introduction to computer graphics: brief history, applications, hardware and software and the fundamental ideas behind modern computer graphics.

Two dimensional graphics: device-independent programming; graphics primitives and attributes.

Interactive graphics: physical input devices, event-driven input; user interface. Transformations; translation, rotation, scaling, shear.

Three-dimensional graphics: 3D curves and surfaces; projections.

Multimedia System Architecture. Objects for Multimedia System: Text; Images and graphics: Basic concepts, Computer image processing; Sound/ Audio: Basic concepts, Music, MIDI, Speech; Video and animation: Basic concepts, Computer-based animation.



Data Compression Techniques: JPEG; H.261 (px64); MPEG; Intel's DVI; Microsoft AVI; Audio compression; Fractal compression Multimedia File Standards: RTF; TIFF; RIFF; MIDI; JPEG DIB; AVI Indeo; MPEG.

Multimedia Storage and Retrieval Technology: Magnetic media technology; Optical media technology: Basic technology, CD Digital audio, CD-ROM, its architecture and further development, CD-Write only (CD-WO), CD- Magnetic optical (CD-MO).

Architecture and Multimedia Communication Systems: Pen input; Video and image display systems; Specialized processors: DSP; Memory systems; Multimedia board solutions; Multimedia communication system; Multimedia database system (MDBMS).

User Interfaces: General design; Video and Audio at the user interface.

Multimedia Applications: Imaging; Image/Voice processing and recognition; Optical character recognition; Communication: Teleservice, Messaging; Entertainment: Virtual reality, Interactive audio and video, Games.

**Recommended Texts:**

1. J. D. Foley et al., *Introduction to Computer Graphics*, 2nd ed. Addison-Wesley Reading, 1994
2. D. Hearn et al., *Computer Graphics with OpenGL*, 4th ed. Pearson, 2010
3. R. Steinmetz and K. Nahrstedt, *Multimedia Systems*. Springer, 2004. DOI: 10.1007/978-3-662-08878-4

**SWE 4632 System Programming and Device Driver Lab Credit: 0.75**

Sessional works based on CSE 4631.

**SWE 4638 Web and Mobile Application Development Lab Credit: 0.75**

Sessional works based on SWE 4637.

The students will incrementally deliver a web and a mobile front-end app that will consume the web service developed in SWE 4537 (Server Programming) course. Some of the features will be available in all platforms, some features will be exclusive to each platform. The requirements of the app will be given week-by-week, the delivery will also be in a weekly interval.

Suggested Technologies — Web: HTML, CSS, Sass/Less, JavaScript, TypeScript; Native Mobile: Android; Cross-platform Mobile: Flutter/Xamarin/Ionic/PhoneGap/React Native; Desktop: Java/C#/Python/Node.